Controlled-NOT (CNOT) Gate

Quick Recap

• A two-qubit system has four basis states: |00>, |01>, |10> and |11>.

• For n qubits, the number of basis states is 2^n.

• Today we learn how one qubit can control an operation on another qubit.

1. What Is the CNOT Gate?

The Controlled-NOT (CNOT or CX) gate is a two-qubit quantum gate. It flips the target qubit only when the control qubit is in state |1>.

• The control decides whether the operation happens.

• The target receives the NOT (X) operation when the control is 1.

• The control qubit is not flipped by the CNOT operation.

Everyday analogy: Think of a switch that enables a lamp: only when the switch is ON does the lamp change its state. This illustrates the condition, not the physics of a qubit.

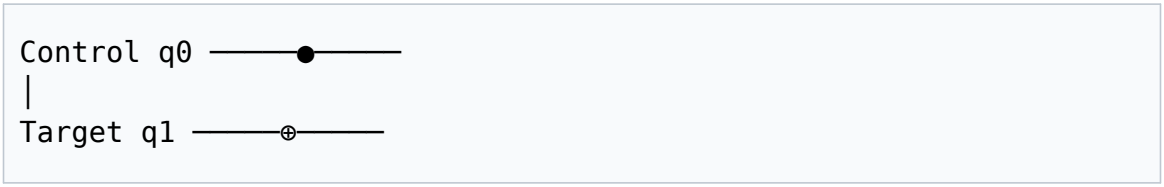


Figure 1. CNOT circuit symbol: filled dot = control; circled plus = target.

2. CNOT Truth Table

Input (control, target)	Output	Explanation
00	00	Control 0: no flip
01	01	Control 0: no flip
10	11	Control 1: flip target
11	10	Control 1: flip target

Example: input |10> becomes |11> because the control is 1 and the target changes from 0 to 1.

3. Why Is It Called Controlled-NOT?

A normal X gate flips one qubit. CNOT applies that flip to the target only when another qubit, the control, is |1>. This makes CNOT a conditional two-qubit gate.

4. Mathematical Representation

In the computational basis ordered |00>, |01>, |10>, |11>, the CNOT operation is represented by a 4 × 4 matrix:

```
[ 1 0 0 0 ]
CNOT = [ 0 1 0 0 ]
[ 0 0 0 1 ]
[ 0 0 1 0 ]
```

Figure 2. The CNOT matrix leaves |00> and |01> unchanged and swaps |10> with |11>.

5. CNOT in Qiskit

The workbook uses `qc.cx(control, target)` to apply CNOT. In the example, q0 controls q1.

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
qc.x(0)                 # Prepare control in |1>
qc.cx(0, 1)             # CNOT: q0 controls q1
qc.measure([0, 1], [0, 1])
print(qc.draw())
```

State progression: |00> → apply X on q0 → |10> → apply CNOT → |11>. Measurement gives the corresponding classical outcome for this prepared basis state.

6. Why CNOT Matters

• Allows interactions between qubits in larger quantum circuits.

• Helps create entangled states when combined with suitable state preparation.

• Appears in quantum teleportation, quantum error correction and quantum algorithms.

Everyday analogy: A classroom rule says, “If the leader raises a hand, the second student switches their sign.” The second student changes only when the condition is met.

7. Common Beginner Mistakes

• “The target always flips.” No. It flips only when the control is 1 for computational basis inputs.

• “Both qubits change.” No. The control remains unchanged in these basis-state examples.

• “CNOT is the same as X.” No. X is a single-qubit gate; CNOT is conditional and acts on two qubits.

Summary: What to Remember

• CNOT means Controlled-NOT and is also written CX.

• It has one control qubit and one target qubit.

• Control 0 → target unchanged; control 1 → target flipped.

• Its computational-basis action is 00→00, 01→01, 10→11, 11→10.

• Qiskit syntax: qc.cx(control, target).

• CNOT is important for entanglement and multi-qubit algorithms.

Resources Mentioned in the Workbook

IBM Quantum CXGate API documentation; Quantum Computing UK CNOT tutorial; Open University Controlled NOT Gate; Computational Mindset NOT and CNOT Operators.